Predicting Emergency Department Crowding: A Markov Chain Model of Hourly NEDOCS States

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Abstract

Emergency Department (ED) overcrowding remains a persistent challenge for hospitals, often driven by a combination of input factors (e.g., rising acuity of patient presentations), throughput issues (e.g., staffing shortages, diagnostic delays), and output constraints (e.g., limited inpatient bed availability). To address the lack of a universal definition for ED overcrowding, the National Emergency Department Overcrowding Score (NEDOCS) was developed in 2004, providing a standardized framework to quantify overcrowding across six severity categories.

This study evaluated the application of a Markov chain model to predict future NEDOCS states, comparing the performance of models using approximately 16 months and 52 months of historical data. Overall, the four-year model demonstrated modestly better predictive performance, especially in accurately forecasting the “Extremely Busy” and “Critical” states. Both models achieved approximately 80% accuracy for one-hour (t+1) forecasts, with performance declining to around 70% at two hours (t+2). Notably, the model's predictions for the "Critical" state remained robust, achieving 87%, 83%, 82%, and 84% accuracy across one- to four-hour forecast horizons. In contrast, predictions for the "Severe" state declined more sharply after t+1.

The results suggest that Markov modeling is a promising approach for short-term NEDOCS forecasting, particularly for the most critical overcrowding states. Given that the models required minimal feature engineering, future work could evaluate whether incorporating additional operational variables improves predictive performance.

Introduction

Hospital Emergency Departments (ED) often contend with various issues that can lead to overcrowding which can have a negative impact on patient care. There are numerous reasons as to why an ED can become overcrowded, but some common issues are "Input" factors such as presentation with more urgent and complex care needs, "Throughput" factors such as staffing shortages or delays in receiving test results or "Output" factors such as lack of available hospital beds (Sartini et al., 2022).

In 2004 a scoring system called NEDOCS (National Emergency Department Overcrowding Score) was created since there was "No single universal definition of emergency department (ED) overcrowding..." (Weiss et al., 2004). Since then, the NEDOCS scoring system has become a common measure to quantify overcrowding by creating a score that stratifies crowding into six ranges as seen in Figure 1.

Figure 1

NEDOCS Levels

Score interpretation:

Level	Score	Interpretation
Level 1	1–20	Not busy
Level 2	21–60	Busy
Level 3	61–100	Extremely busy but not overcrowded
Level 4	101–140	Overcrowded
Level 5	141–180	Severely overcrowded
Level 6	181–200	Dangerously overcrowded

Markov chain methods have previously been applied to emergency department forecasting. Zhang et al. (2013) proposed a Markov chain model to forecast patient flow between different stages of the emergency department process. The current study applies a similar state-transition approach to a different forecasting problem by modeling transitions between hourly NEDOCS crowding states.

The NEDOCS scoring system is great for indicating current levels, however it is not a forecasting model and therefore has limited predictive power. The purpose of this research is to use a Markov chain model based on historical NEDOCS scores to predict future state NEDOCS levels at various hourly time-steps.

Methodology

Data was acquired from a busy Emergency Department of modest size that treats ~63,000 patients a year. The data represents hourly NEDOCS scores over a four-and-a-half-year period and was divided into two analysis periods consisting of approximately 16 months and 52 months of data. Each dataset was then divided using an 80/20 train-test split. For consistency throughout this paper, the shorter training period is referred to as the "one-year" model and the longer period as the "four-year" model. The training sets were used to generate time-dependent Markov transition matrices that represent each hour of each day of the week. This resulted in 168 (7 days x 24 hours) transition matrices for each training set (an example can be seen in Fig. 2). The transition probabilities were calculated as follows:

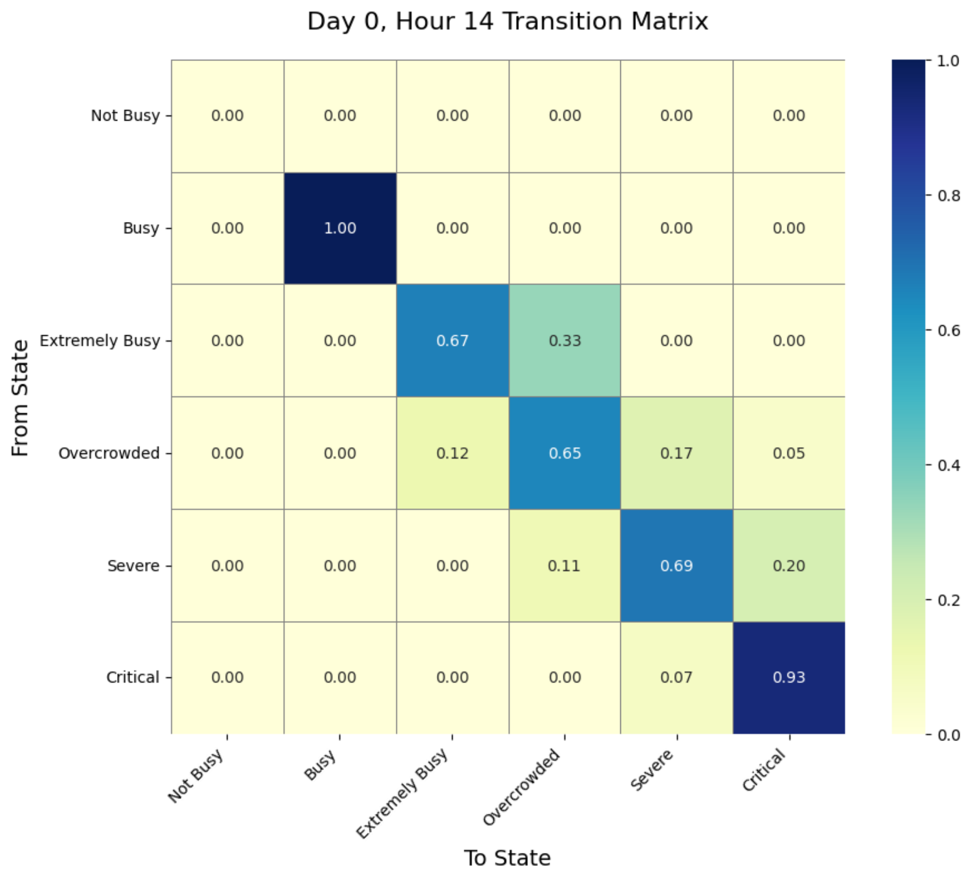
$$P_{\{ij\}}^{\{(d,h)\}} = Pr\left(X_{\{t+1\}} = j \mid X_t = i, Day = d, Hour = h\right)$$

- (X_t) $is\ the\ system\ state\ at\ time\ (t)$
- $d \in \{0, 1, \ldots, 6\})$ $represents\ the\ day\ of\ the\ week$
- $h \in \{0, 1, \ldots, 23\})$ $represents\ the\ hour\ of\ the\ day$
- $P^{\{(d,h)\}\}} is\ a\ distinct\ transition\ matrix\ for\ each\ ((d,h))\ pair$

Each test set was then evaluated against the trained Markov matrices. Forecast accuracy was calculated using a deterministic process that selected the state with the highest predicted probability. Some of the NEDOCS states were renamed for ease of formatting, specifically: "Extremely Busy but not overcrowded" was renamed to "Extremely Busy", "Severely Overcrowded" to "Severe" and "Dangerously Overcrowded" to "Critical".

Figure 2

Sample Transition Matrix

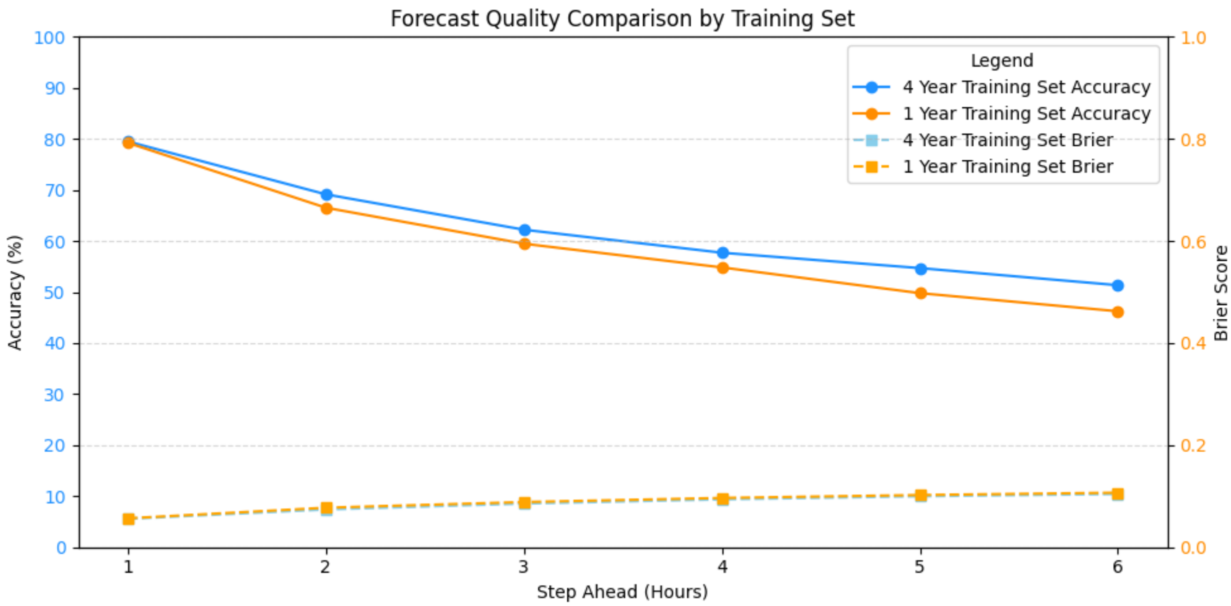


Results and Discussion

The model was evaluated against the 20% test sets and predictions were made up to six hours into the future. The predicted NEDOCS states were then compared to the true states and an overall percentage score calculated for each time step. A Brier score was also calculated to assess the accuracy of the probabilistic predictions, providing a measure of how closely the predicted probabilities aligned with the observed outcomes. As seen in Figure 3, overall accuracy for predictions one hour in advance is around 80% with a steady decrease in accuracy for each additional hour. The Brier score remained well below 0.2 and increased as the forecast horizon extended, reflecting increasing uncertainty in longer-range predictions.

Figure 3

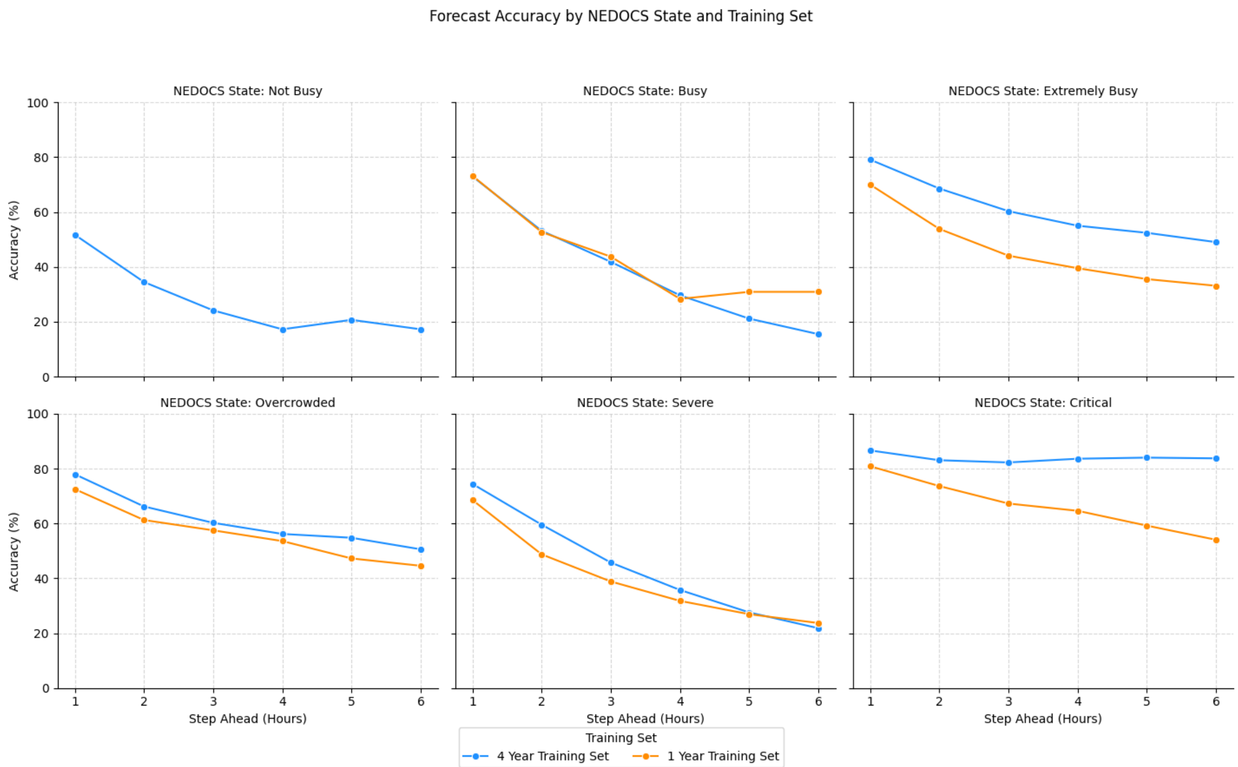
Overall forecast accuracy



Accuracy was then graphed by each NEDOCS state for each time step in Figure 4. The four-year model generally performed better than the one-year model, with the largest differences occurring in the Extremely Busy and Critical states. The Critical state also maintained the highest accuracy across the forecast horizon.

Figure 4

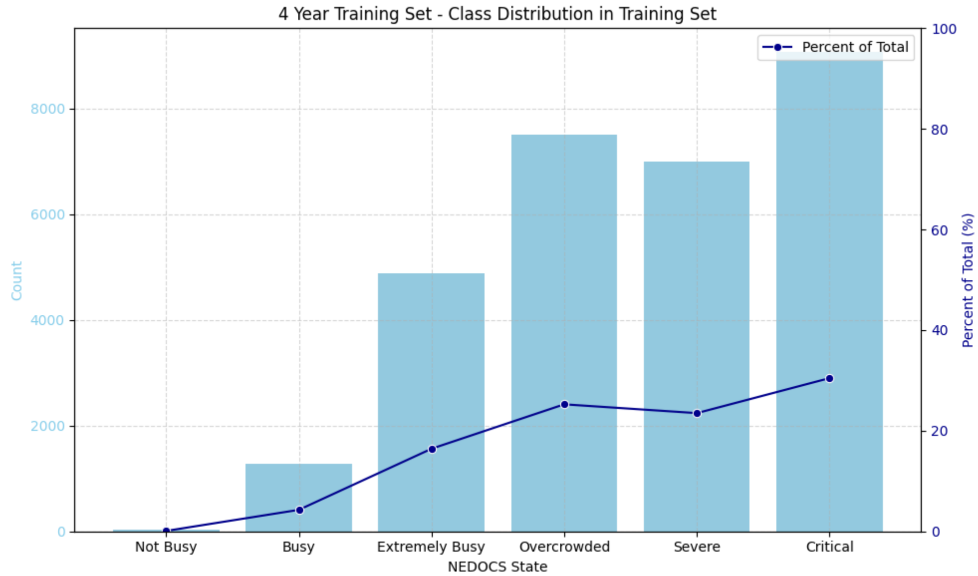
NEDOCS State Accuracy



When evaluating the distribution of NEDOCS observations, a substantial class imbalance is evident, with observations concentrated in the higher crowding states (Fig. 5). The Critical state represents the largest portion of the training data, while the Not Busy state is rarely observed. This imbalance may contribute to differences in predictive performance across NEDOCS states, particularly the model's relatively strong performance when forecasting the Critical state.

Figure 5

Distribution of NEDOCS States in the Four-Year Training Set



When predictions were incorrect, misclassifications tended to occur among neighboring NEDOCS states. The normalized confusion matrices (Fig. 6) provide additional detail about how predictions were distributed across NEDOCS states. Diagonal values represent correct classifications, while off-diagonal values show how incorrect predictions were distributed among the remaining states. For example, at time step t+1, when the actual state was Severe, the model predicted Extremely Busy only 0.2% of the time, compared with Overcrowded (13.6%) and Critical (11.9%), both of which are adjacent to the true state. A similar pattern of misclassification is observed in the one-year model, as shown in Figure 7.

Figure 6

Four-Year Training Model Normalized Confusion Matrix

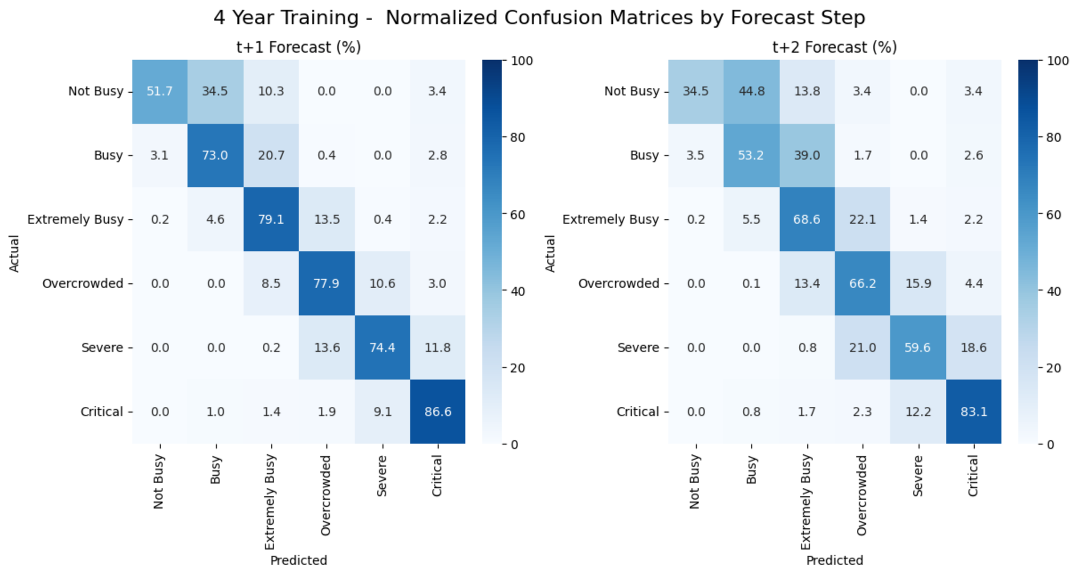
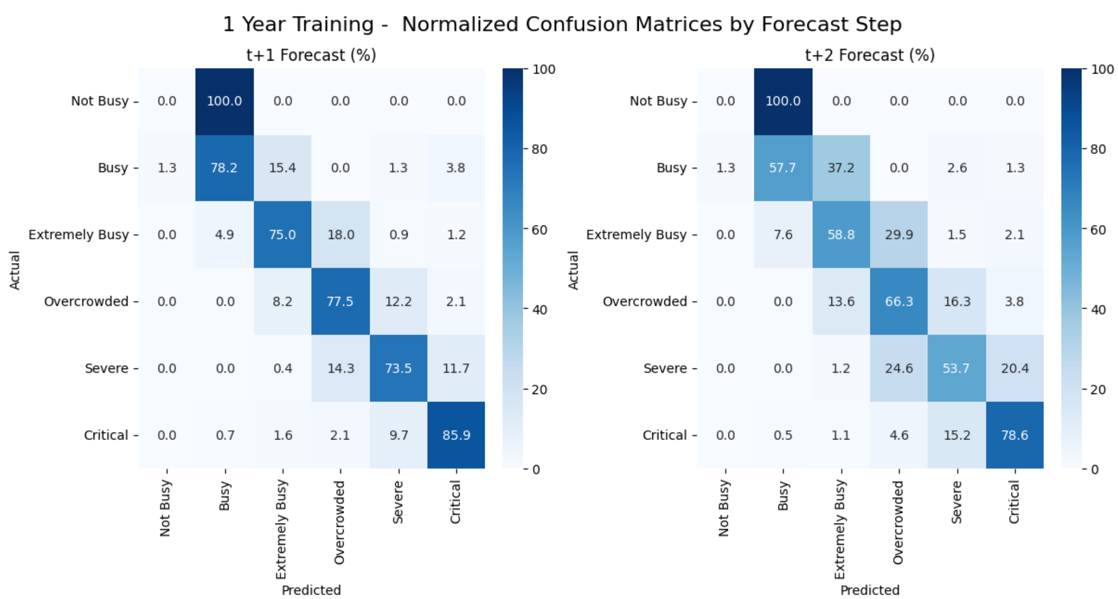


Figure 7

One-Year Training Model Normalized Confusion Matrix



Conclusion

This study demonstrated that a time-dependent Markov chain model can provide useful short-term forecasts of emergency department crowding based on historical NEDOCS states. Both models achieved approximately 80% accuracy at the one-hour forecast horizon, with accuracy declining as the forecast period increased. The longer historical training period generally produced modest improvements in overall performance, although the benefit varied substantially across individual NEDOCS states. The largest differences were observed for the Extremely Busy and Critical states, while other states showed more comparable performance between the two models.

The analysis also identified important limitations. NEDOCS states were not evenly represented in the historical data, with substantially more observations occurring in the higher crowding states. This imbalance may have contributed to differences in state-level forecast performance, although the analysis does not establish a causal relationship.

Overall, the results suggest that Markov chain modeling may be useful as a relatively simple approach for short-term NEDOCS forecasting. Future work could evaluate whether incorporating additional operational variables, addressing sparse state-transition combinations, and comparing the Markov approach with other statistical or machine-learning methods provide meaningful improvements in predictive performance.

References

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